

Universality and the right-angled Coxeter envelope of orthoscheme reflection groups

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Abstract

The right-triangle orbit-growth universality of the companion papers is the planar case of a general phenomenon. We prove an abstract dichotomy: for any family of quotients of a common group, spherical growth is shape-independent through a horizon equal to half the shortest added relator and first deviates exactly there. For orthoscheme (path-simplex) reflection groups the common group is the *right-angled Coxeter group* (RACG) on the orthoscheme's orthogonality graph, whose growth we compute in closed form: it is rational, with rate the algebraic number $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$ (the spectral radius attached to the path P_{n+1}) and numerator $(1+t)^{\lfloor n/2 \rfloor + 1}$. The actual geometric group $W_T \leq \text{Isom}(\mathbb{R}^n)$ is *amenable*, hence a *proper* quotient of the non-amenable RACG, so its growth matches this rational envelope only up to a finite, shape-dependent depth and then deviates strictly below. In the plane the deviation occurs at depth 10, where the growth becomes the transcendental sequence A396406; in every higher dimension it lies beyond direct computation. We are therefore *not* entitled to conclude that higher dimensions are rational: the finite envelope agreement observed for $n \geq 3$ is the same pattern that, at $n = 2$, precedes the transcendental collapse. Whether the plane is genuinely exceptional, or every dimension is eventually transcendental, is left as an explicit open problem.

1 The abstract universality–deviation dichotomy

Let $G = \langle S \rangle$ be a group with finite symmetric generating set S and word length ℓ_G . For $N \trianglelefteq G$ write $Q = G/N$, $s_d(H) = \#\{h : \ell_H(h) = d\}$, and $K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$.

Theorem 1 (Universality–deviation dichotomy). *With $n := \lceil K(N)/2 \rceil$: $s_d(Q) = s_d(G)$ for $d < n$, and $s_n(Q) < s_n(G)$.*

Proof. $\pi : G \rightarrow Q$ is surjective and 1-Lipschitz, so $s_d(Q) \leq s_d(G)$. If $d < n$ and $\pi g = \pi h$ ($g, h \in B_d(G)$) then $gh^{-1} \in N$ has length $\leq 2d < K(N)$, forcing $g = h$; the same bound excludes a shorter representative, so π is length-preserving on $B_d(G)$ and $s_d(Q) = s_d(G)$ for $d < n$. Splitting a geodesic word for $g_0 \in N$ with $\ell_G(g_0) = K(N)$ as $g_0 = w_1 w_2$, $\ell_G(w_1) = n$, gives two distinct elements of $B_n(G)$ sharing a fiber, so $s_n(Q) < s_n(G)$. $\square$

Applied to a family $\{Q_T = G/N_T\}$, this is universality: all members share G 's growth through their horizon $n_T = \lceil K(N_T)/2 \rceil$, then deviate. **The point of this paper is that for orthoscheme reflection groups the ambient G is a rational-growth RACG, but the members Q_T are proper quotients, so the shared rational growth is only a finite-depth envelope.**

2 The orthoscheme family and its RACG envelope

An n -orthoscheme has vertices $P_0, \dots, P_n$ with mutually orthogonal consecutive edges $P_{k-1}P_k = \ell_k e_k$; its $n+1$ facet reflections $R_0, \dots, R_n$ generate $W_T \leq \text{Isom}(\mathbb{R}^n)$ (the right-triangle group when

$n = 2$). Two facts hold for *every* orthoscheme: $R_i^2 = 1$, and $R_i R_j = R_j R_i$ when $|i - j| \geq 2$ (orthogonal facets). For generic legs consecutive facets meet at an irrational multiple of π , imposing no finite bond. The largest group compatible with these relations is the *right-angled Coxeter group*

$$W_n = \langle R_0, \dots, R_n \mid R_i^2 = 1, (R_i R_j)^2 = 1 \ (|i - j| \geq 2) \rangle,$$

and there is a surjection $\rho: W_n \twoheadrightarrow W_T$.

Before computing the envelope we record that the “generic growth” is a well-defined object: it does not depend on which generic shape one picks. Let $F = \ast_{i=0}^n \langle r_i \rangle$ be the free product of $n + 1$ copies of $\mathbb{Z}/2$, let $U = \{\ell = (\ell_1, \dots, \ell_n) : \ell_i > 0\}$ be the leg space ($\ell_1 = 1$), and let $\rho_\ell: F \rightarrow \text{Isom}(\mathbb{R}^n)$, $r_i \mapsto R_i(\ell)$, be the reflection representation; put $N_{\text{gen}} = \bigcap_{\ell \in U} \ker \rho_\ell$.

Theorem 2 (Generic rigidity). *For every $n \geq 2$ there is a co-null subset $U_{\text{gen}} \subseteq U$ — the complement of a countable union of proper real-algebraic subvarieties — such that $\ker \rho_\ell = N_{\text{gen}}$, hence $W_\ell \cong F/N_{\text{gen}}$, for every $\ell \in U_{\text{gen}}$. Consequently the geodesic growth $s_d(W_\ell)$ is constant on U_{gen} , equal to $s_d(F/N_{\text{gen}})$, and this generic value is the maximum of $s_d(W_\ell)$ over all $\ell \in U$. Moreover U_{gen} contains every ℓ whose coordinates are algebraically independent over $\mathbb{Q}$; in particular it is dense.*

Proof. Each facet normal $m_j(\ell)$ is a fixed polynomial vector in ℓ (a cofactor of the vertex differences, which are linear in ℓ), so the reflection $R_j(\ell) = I - 2m_j m_j^\top / (m_j \cdot m_j)$ (with its translation) has entries in $\mathbb{Q}(\ell)$ with denominator the nonvanishing polynomial $m_j \cdot m_j$. Hence every entry of $\rho_\ell(v) = \prod_k R_{i_k}(\ell)$ lies in $\mathbb{Q}(\ell)$, and for each $v \in F$ the locus $V_v = \{\ell \in U : \rho_\ell(v) = I\}$ — the simultaneous vanishing of finitely many rational functions, cleared of their nonvanishing denominators — is Zariski-closed in the irreducible variety U . If $v \notin N_{\text{gen}}$ then $\rho_\ell(v) \neq I$ for some ℓ , so V_v is a *proper* closed subvariety, of measure zero. As F is countable, $B := \bigcup_{v \in F \setminus N_{\text{gen}}} V_v$ is a countable union of proper subvarieties; $U_{\text{gen}} := U \setminus B$ is co-null and dense. For $\ell \in U_{\text{gen}}$ one has $N_{\text{gen}} \subseteq \ker \rho_\ell$ (universal relations hold everywhere) and $\ker \rho_\ell \subseteq N_{\text{gen}}$ (any $v \in \ker \rho_\ell \setminus N_{\text{gen}}$ would place $\ell \in V_v \subseteq B$), so $\ker \rho_\ell = N_{\text{gen}}$ and $W_\ell \cong F/N_{\text{gen}}$. Isomorphic groups with the marked generators have identical geodesic growth, giving the constant value; for any ℓ , $\ker \rho_\ell \supseteq N_{\text{gen}}$ makes W_ℓ a quotient of F/N_{gen} , so $s_d(W_\ell) \leq s_d(F/N_{\text{gen}})$, i.e. the generic value is the maximum. Finally each V_v is defined over $\mathbb{Q}$; if the coordinates of ℓ are algebraically independent over $\mathbb{Q}$ then ℓ lies on no proper $\mathbb{Q}$ -subvariety, so $\ell \notin B$, i.e. $\ell \in U_{\text{gen}}$. Such ℓ exist since $\dim U = n - 1 \geq 1$. $\square$

The non-consecutive commuting relations $(R_i R_j)^2 = 1$ ($|i - j| \geq 2$) hold identically in ℓ (the interior normals m_j are supported on the two coordinates $\{j-1, j\}$, so non-adjacent normals are orthogonal for *all* legs; verified symbolically for $n \leq 5$), so they lie in N_{gen} and F/N_{gen} is a quotient of the RACG W_n below.

Theorem 3 (The RACG envelope is rational). *The spherical growth series of W_n is rational,*

$$W_n(t) = \frac{(1+t)^{\lfloor n/2 \rfloor + 1}}{N_n(t)}, \quad N_n(t) = (1+t)^{n+1} c_{\Gamma_n} \left(\frac{-1}{1+t} \right),$$

where c_{Γ_n} is the clique polynomial of the orthogonality graph Γ_n (vertices $0, \dots, n$; edges $|i - j| \geq 2$). Its growth rate is $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$, the largest root of the reciprocal of N_n , and equals 1 plus the

spectral radius $2 \cos \frac{2\pi}{n+3}$ of the path P_{n+1} . Explicitly

n	$W_n(t)$	r_n
2	$(1+t)^2/(1-t-t^2)$	$\varphi = 1.618\dots$
3	$(1+t)^2/(1-2t)$	2
4	$(1+t)^3/(1-2t-t^2+t^3)$	$1 + 2 \cos \frac{2\pi}{7} = 2.24698\dots$
5	$(1+t)^3/(1-3t+t^2+t^3)$	$1 + \sqrt{2}$
6	$(1+t)^4/(1-3t+3t^3)$	$1 + 2 \cos \frac{2\pi}{9} = 2.53209\dots$
7	$(1+t)^4/(1-2t-3t^2+4t^3-t^4)$	$\varphi^2 = 2.61803\dots$

Proof. W_n is a RACG, so its finite parabolics are exactly the cliques of Γ_n , each a group $(\mathbb{Z}/2)^{|C|}$ with polynomial growth $(1+t)^{|C|}$. Steinberg’s formula gives $1/W_n(t^{-1}) = \sum_C \text{clique}(-1)^{|C|}/(1+t)^{|C|} = c_{\Gamma_n}(-1/(1+t))$; inverting and substituting $t \mapsto t^{-1}$ yields the displayed rational form. Γ_n is the complement of the path P_{n+1} , and the reciprocal of N_n is (a Chebyshev multiple of) the characteristic polynomial of P_{n+1} , whose largest eigenvalue is $2 \cos \frac{\pi}{n+2}$; a short computation with the independence polynomial of the path pins the dominant growth root at $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$. The tabulated entries are the cases $n = 2, \dots, 7$, verified symbolically. $\square$

3 The geometric group is a proper amenable quotient

Proposition 4 (Finite-horizon envelope). *For every $n \geq 2$, W_T is a proper quotient of W_n ; consequently, by Theorem 1, the geometric growth $s_d(W_T)$ equals the rational envelope $[t^d]W_n(t)$ only for $d < n_T := \lceil K(\ker \rho)/2 \rceil$, and $s_{n_T}(W_T) < [t^{n_T}]W_n(t)$.*

Proof. $W_T \leq \text{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$ is amenable (an extension of the amenable $\mathbb{R}^n$ by the compact $O(n)$). The RACG W_n is non-amenable: its orthogonality graph Γ_n is not complete multipartite (its complement, the path P_{n+1} , is connected and not a disjoint union of cliques for $n \geq 2$), so W_n contains a free subgroup of rank 2. An amenable group cannot be isomorphic to a non-amenable one, so $\ker \rho \neq \{1\}$ and W_T is a proper quotient. Theorem 1 then gives the horizon. $\square$

Corollary 5 (The plane). *For $n = 2$ the horizon is $n_T = 10$: $s_d(W_T) = F_{d+3} = [t^d]W_2(t)$ for $d \leq 9$, and at $d = 10$ the growth deviates, becoming the sequence A396406, which is transcendental over $\mathbb{Q}$ (companion paper: the relaxed companion unconditionally; the true series conditionally on one amplitude estimate) with rate $\beta_2 = 1.4916\dots < \varphi = r_2$. This is the only dimension in which the deviation has been computed past its onset.*

4 What is proved, and the open problem

Four things are proved unconditionally and dimension-freely: the universality–deviation dichotomy (Theorem 1); *generic rigidity* (Theorem 2), which makes “the generic n -orthoscheme growth” a well-defined object $s_d(F/N_{\text{gen}})$ realized by a co-null set of shapes; the rationality of the RACG envelope with rate r_n (Theorem 3); and, structurally, that the geometric group is always a proper amenable quotient, so its growth is the envelope only up to a finite horizon (Proposition 4). For $3 \leq n \leq 7$ the geometric growth was computed to depth 13–16 and agrees with the envelope throughout that

range; but Proposition 4 guarantees it must eventually deviate below, and the deviation depth n_T for $n \geq 3$ exceeds what we can reach.

The tempting conclusion — that the geometric growth is rational for $n \geq 3$, so the plane is the unique transcendental dimension — is *not* justified. The evidence for it is precisely the finite envelope agreement, and at $n = 2$ that same agreement (through depth 9) precedes the transcendental collapse. We therefore state the honest alternative as an open problem rather than a theorem.

Question 6 (Eventual transcendence). For $n \geq 3$, is the geometric growth series of the generic n -orthoscheme group rational, algebraic, or transcendental? Equivalently: what is the arithmetic of the amenable quotient $W_n/\ker \rho$ past its deviation horizon n_T ? The plane is exceptional iff the answer is “rational” for all $n \geq 3$; the alternative — eventual transcendence in every dimension, with the plane merely the earliest and only computable case — is equally consistent with all current evidence.

The companion papers are recast accordingly: their object, the planar right-triangle group, is the one dimension in which the amenable quotient of the Coxeter envelope has been carried past its deviation and found transcendental. Whether that transcendence is a two-dimensional accident or the first instance of a universal phenomenon is open.